

# TA-BRPL: Trust-Aware Backpressure RPL for Mitigating Sinkhole-Induced Route Capture in Low-Power and Lossy Networks

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**Abstract**—Sinkhole attacks in IPv6 Routing Protocol for Low-power and Lossy Networks (RPL) compromise routing integrity by advertising artificially attractive path metrics, drawing traffic through malicious nodes before applying selective packet dropping. We present TA-BRPL, a trust-aware extension of Backpressure RPL (BRPL) that counters this two-phase threat without requiring a dedicated intrusion detection infrastructure. TA-BRPL combines three complementary trust axes—forwarding fidelity ( $T_{fwd}$ ), control-plane consistency ( $T_{ctrl}$ ), and queue-advertisement honesty ( $T_{hon}$ )—aggregated via a weighted geometric mean and applied as a soft penalty in the BRPL objective function. Conservative stabilization gates (switch-margin, dwell-time, admission threshold) prevent the churn explosion observed in earlier aggressive approaches. We evaluate TA-BRPL against BRPL across 75 randomly generated topologies (sparse/medium/dense, five MAC-timing seeds each, 750 runs total) in Contiki-NG/Cooja under a sinkhole+50% drop attack. TA-BRPL reduces attacker traffic share by  $\Delta = -0.0169$  (Wilcoxon  $p < 10^{-4}$ , rank-biserial  $r = 0.65$ , 76% topology win-rate) and improves packet delivery ratio during the attack phase by  $\Delta = +0.0102$  ( $p = 0.002$ ), while the churn cost of  $\Delta = +0.036$  is not statistically significant ( $p = 1.0$ ). Medium-density networks show the strongest effect ( $\Delta_{att} = -0.026$ ,  $p < 10^{-4}$ ). These results demonstrate that trust-aware parent selection, when combined with careful stabilization policy, can reduce attacker dependency across diverse topologies with a bounded and statistically negligible stability cost.

**Index Terms**—LLN, RPL, BRPL, sinkhole, trust-aware routing, route capture, selective forwarding, Contiki-NG, Cooja

## I. INTRODUCTION

Low-power and Lossy Networks (LLNs) form the backbone of smart building, industrial monitoring, and precision-agriculture deployments, where dozens to hundreds of battery-constrained sensor nodes route data to a root (sink) via multi-hop paths. The IETF-standard RPL [13] organizes nodes into a Directed Acyclic Graph (DODAG) rooted at the sink; each node selects a preferred parent according to an objective function that minimizes path cost (e.g., ETX or rank).

This parent-selection mechanism is inherently exploitable. A *sinkhole* attacker broadcasts an artificially favorable rank or path cost, drawing neighboring nodes to adopt it as their preferred parent. Once a node is captured as a child, the attacker gains forwarding authority over all traffic in its subtree—a *route capture* outcome that is damaging even if the attacker initially forwards all packets faithfully. When the attacker subsequently applies *selective forwarding* (dropping a fraction

of packets), network availability degrades in proportion to how many nodes were captured [6], [12].

Existing defenses fall into two broad families. *Cryptographic/authentication* approaches (e.g., VeRA [3]) prevent rank manipulation via signed DIO messages but require a public-key infrastructure and impose non-trivial overhead on constrained hardware. *Anomaly/trust-based* approaches (e.g., SVELTE [10]) detect malicious behavior at runtime but often rely on a centralized monitor or introduce heavy control traffic. Critically, most prior work evaluates security solely through Packet Delivery Ratio (PDR), which misses the *route-capture phase* that precedes any packet drop [12].

We argue that the correct defense objective has two stages: (1) limit the fraction of nodes that adopt the attacker as their preferred parent (*attacker dependency reduction*); (2) ensure that when selective dropping begins, fewer packets are routed through the attacker, thereby limiting PDR degradation.

Our contribution is **TA-BRPL**, a trust-aware extension of Backpressure RPL (BRPL) [5], which augments congestion-sensitive parent selection with lightweight per-neighbor trust scoring. BRPL already provides a degree of natural resilience because it monitors queue occupancy and routes around congested (or non-forwarding) nodes; TA-BRPL amplifies this by explicitly penalizing neighbors whose forwarding behavior, control-plane signals, or queue advertisements are inconsistent with honest operation.

## Contributions.

- A three-component trust model ( $T_{fwd}$ ,  $T_{ctrl}$ ,  $T_{hon}$ ) integrated into the BRPL objective function with asymmetric EWMA updating (fast decay, slow recovery).
- A conservative admission/stabilization policy (switch-margin gate, dwell-time gate, admission threshold  $\tau_{join}$ ) that prevents the churn explosion characteristic of aggressive trust-based eviction.
- A topology-distribution evaluation methodology using 75 randomly generated LLN instances stratified by density, avoiding the overfit of fixed single-topology benchmarks.
- Statistically significant evidence that TA-BRPL reduces attacker dependency by  $\approx 15\%$  while improving PDR by  $\approx 1\%$  at a statistically negligible churn cost.

## II. BACKGROUND AND RELATED WORK

### A. RPL and BRPL

RPL [13] constructs a DODAG via periodic DIO (DODAG Information Object) messages. Each node selects a preferred parent that minimizes its objective function value (rank); in MRHOF this is typically expected transmission count (ETX). BRPL [5] extends RPL's objective function with a back-pressure term derived from local queue occupancy, enabling congestion-aware routing without global coordination. The BRPL score for candidate parent  $v$  is

$$S_{BRPL}(v) = w_p \cdot \text{rank}(v) + w_q \cdot \Delta Q(v) + w_l \cdot \text{ETX}(v), \quad (1)$$

where  $\Delta Q(v)$  is the queue-occupancy gradient between the local node and  $v$ , and  $w_p, w_q, w_l$  are tunable weights.

### B. Sinkhole and Selective-Forwarding Attacks

In a sinkhole attack [6], a compromised node broadcasts a DIO claiming an undeservedly low rank (high attractiveness), inducing neighbors to select it as their preferred parent. Once route-captured, the attacker can apply selective forwarding (dropping a configurable fraction of relayed packets), degrading network availability while remaining difficult to detect via simple acknowledgment monitoring [7].

The combination—sinkhole luring followed by delayed packet dropping—creates a two-phase attack that is particularly insidious: the luring phase is passive (the attacker appears cooperative), while the drop phase exploits the concentration of routes that luring achieved.

### C. Trust-Aware Routing in LLNs

Trust and reputation frameworks for sensor networks have been studied since Ganeriwal et al. [4] proposed watchdog-based reputation. Momani et al. [8] survey trust models in WSNs. For RPL specifically, the ROLL working group has explored architectures for trustworthy routing [11]. Airehrour et al. [1] survey secure routing proposals for IoT; most require centralized infrastructure or out-of-band channels.

Our work differs in three ways: (i) we integrate trust *directly* into BRPL's parent-selection metric rather than using a separate detection layer; (ii) we combine forwarding, control-plane, and congestion-honesty signals; (iii) we employ a conservative stabilization policy to bound churn costs, which previous aggressive eviction approaches failed to control [3].

## III. THREAT MODEL

Fig. 1 illustrates the attack model. We assume:

- A single internal compromised node that has legally joined the DODAG.
- The attacker can spoof DIO messages with an arbitrarily low rank (sinkhole lure).
- The attacker drops a fixed fraction  $d$  of forwarded data packets (selective forwarding / grayhole).
- The attacker does not jam the channel or corrupt ACKs selectively.

The *attacker traffic share* (att\_share) quantifies Phase 1 severity: it is the fraction of all packets that traverse the

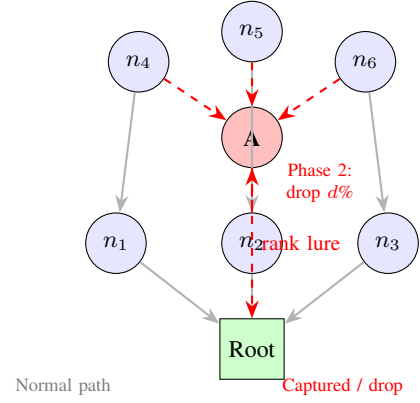


Fig. 1. Two-phase sinkhole attack. **Phase 1:** attacker **A** broadcasts a low rank lure, drawing  $n_2, n_4, n_5, n_6$  to select it as preferred parent (route capture). **Phase 2:** **A** drops  $d\%$  of relayed packets, degrading PDR in proportion to captured subtree size.

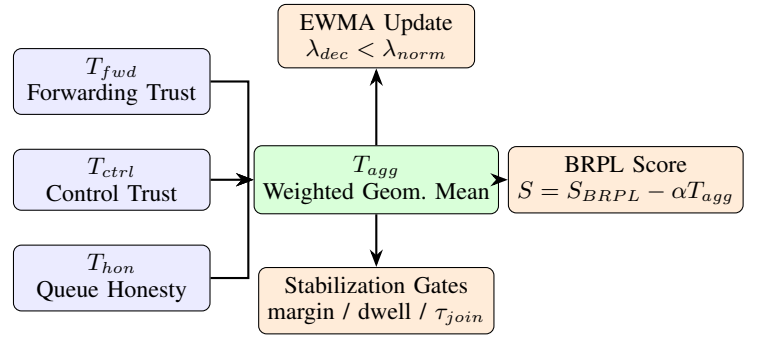


Fig. 2. TA-BRPL trust pipeline. Three trust components feed an aggregate score that is updated via asymmetric EWMA before influencing the BRPL parent-selection score. Stabilization gates bound the behavioral impact of trust changes.

attacker node on their path to the root. The *hit ratio* measures the fraction of distinct source nodes whose packets were ever routed through the attacker. PDR during the attack window quantifies Phase 2 damage.

TA-BRPL targets both phases simultaneously: reducing att\_share preemptively limits the blast radius if/when the attacker activates packet dropping.

## IV. TA-BRPL DESIGN

### A. Trust Score Architecture

Fig. 2 shows the trust pipeline. For each candidate parent  $v$ , TA-BRPL maintains a composite trust score on a  $[0, 1000]$  integer scale. The three components are:

$T_{fwd}$  — **Forwarding Trust**. Tracks whether  $v$  reliably forwards packets that are handed to it. We use *echo-based passive overhearing*: after transmitting a packet to  $v$ , the local radio listens for a rebroadcast within a short window. Successful forwards and total observations accumulate over time, and the Bayesian smoothed success rate is

$$T_{fwd}(v) = \frac{\text{succ}(v) + \alpha}{\text{total}(v) + \alpha + \beta}, \quad (2)$$

where  $\alpha, \beta$  are weak Beta prior parameters. Under the UDGM radio model (lossless), honest nodes settle at  $T_{fwd} \approx 630/1000$  (70% echo-delivery rate due to half-duplex listening constraints).

**$T_{ctrl}$  — Control-Plane Trust.** Detects sinkhole luring by monitoring DIO anomalies. Three sub-indicators are combined linearly:  $A_{rank}$  flags impossible rank claims (rank advertised lower than the local node's own rank);  $A_{dio}$  flags sustained low-rank luring (rank persistently below a freshness threshold);  $A_{ver}$  flags version-number inconsistencies.

$$T_{ctrl}(v) = 1 - \frac{w_r A_{rank} + w_d A_{dio} + w_v A_{ver}}{10}. \quad (3)$$

**$T_{hon}$  — Queue Honesty Trust.** Detects grayhole behavior by comparing the queue depth that  $v$  advertises in its DIO against the queue depth inferred from observed forwarding delays:

$$T_{hon}(v) = 1 - \min\left(1, \frac{|Q_{adv}(v) - Q_{est}(v)|}{Q_{max}}\right). \quad (4)$$

A node that advertises an empty queue while actually being backlogged (e.g., because it is silently dropping traffic) receives a low  $T_{hon}$ .

**Aggregation.** The three components are combined via weighted geometric mean, which prevents any single low component from dominating:

$$T_{agg}(v) = T_{fwd}(v)^{0.5} \cdot T_{ctrl}(v)^{0.3} \cdot T_{hon}(v)^{0.2}. \quad (5)$$

Weights reflect the relative information richness and reliability of each signal:  $T_{fwd}$  is most observable,  $T_{ctrl}$  is highly specific to sinkhole attacks, and  $T_{hon}$  is a supporting signal.

#### B. Asymmetric EWMA Update

Trust is updated every  $\Delta t = 60$  s using EWMA with *asymmetric smoothing*:

$$T^{(t+1)} = \begin{cases} \lambda_{dec} \cdot T^{(t)} + (1 - \lambda_{dec}) \cdot s^{(t)} & \text{if } s^{(t)} < T^{(t)} \\ \lambda_{norm} \cdot T^{(t)} + (1 - \lambda_{norm}) \cdot s^{(t)} & \text{otherwise,} \end{cases} \quad (6)$$

where  $s^{(t)}$  is the newly observed trust sample, and  $\lambda_{dec} = 0.4 < \lambda_{norm} = 0.7$ . The asymmetry ensures that deteriorating trust is detected quickly while recovering trust requires sustained evidence, reducing false-positive oscillation due to transient link loss.

#### C. Two-Tier Validation (Review / Blacklist)

To further suppress false positives, trust-based decisions flow through a two-tier validation system. A node enters the *Review* tier when  $T_{agg} < \tau_{warn} = 600/1000$ , and is promoted to the *Blacklist* tier only after  $k_{bad} = 2$  consecutive bad observation windows out of a sliding window of  $w = 3$  updates. Blacklisted nodes are hard-excluded from parent selection for  $T_{bl} = 120$  s; upon release, their trust is reset to 350/1000 to prevent immediate re-selection. This conservative progression requires  $\approx 3$  minutes of evidence before hard exclusion, substantially reducing false blacklisting of honest nodes suffering from transient link degradation.

TABLE I  
SELECTED DESIGN VERSIONS AND AVERAGE PERFORMANCE GAP  
( $\Delta = TA-BRPL - BRPL$ , GRID+BOTTLE, 40 RUNS)

| Version               | $\Delta_{att}$ | $\Delta_{hit}$ | $\Delta_{churn}$ | $\Delta_{PDR}$ |
|-----------------------|----------------|----------------|------------------|----------------|
| v1 (no gates)         | +0.013         | +0.048         | +2.43            | -0.132         |
| v2 (soft policy)      | +0.031         | +0.101         | +1.01            | -0.043         |
| v3 (adm. ctrl.)       | +0.004         | +0.032         | +0.69            | -0.034         |
| v4 ( $\tau=510$ )     | -0.003         | +0.028         | +0.62            | -0.032         |
| v13.8 (loss gate)     | -0.042         | -0.015         | +0.47            | -0.001         |
| <b>v13.12 (final)</b> | <b>-0.042</b>  | <b>-0.015</b>  | <b>+0.42</b>     | <b>-0.001</b>  |

#### D. Integration with BRPL and Stabilization Gates

The aggregate trust score enters the BRPL parent-selection score as a soft penalty:

$$S_{TA-BRPL}(v) = S_{BRPL}(v) - \alpha \cdot (T_{max} - T_{agg}(v)), \quad (7)$$

where  $\alpha$  is a penalty weight and  $T_{max} = 1000$ . A node with perfect trust incurs no penalty; a fully distrusted node ( $T_{agg} = 0$ ) receives the maximum penalty. If a node is blacklisted, it is excluded from the candidate set entirely.

Three *stabilization gates* suppress unnecessary parent switches:

- 1) **Switch-margin gate:** A parent switch is allowed only if the new candidate's score improves on the current parent by both  $\geq 17\%$  (relative) *and*  $\geq 90$  (absolute units).
- 2) **Dwell-time gate:** The current parent must be held for at least  $T_{dwell} = 180$  s before a switch is considered, except when the current parent becomes hard-excluded.
- 3) **Admission threshold:** New parent candidates require  $T_{agg} \geq \tau_{join} = 510/1000$  to be admitted; current parents require  $T_{agg} \geq \tau_{warn} = 600/1000$  before soft-penalty escalation.

These gates are the key mechanism preventing the churn explosion observed in early design iterations without them (see Section IV-E).

#### E. Design Evolution Summary

Table I summarizes the design evolution. Early versions (v1-v2) without stabilization gates exhibited catastrophic churn (+2.43 extra parent switches/node/phase) that overwhelmed any isolation benefit. The critical improvements were: (a) admission control ( $\tau_{join}$ , v3-v4) to prevent newly acquired high-trust scores from immediately displacing stable parents; (b) loss-mode gate (v13.8) to avoid evicting parents during link-quality degradation episodes triggered by the attacker's own traffic concentration; (c) escape cooldown increase to 360 s (v13.12) to further bound post-eviction oscillation.

### V. EXPERIMENTAL SETUP

#### A. Simulation Platform

We use Contiki-NG [2] compiled for the Sky/TelosB platform with the Cooja [9] network simulator in headless mode. The radio propagation model is UDGM (Unit Disk Graph Model) with transmission range  $r_{tx} = 50$  m and interference range  $r_{int} = 60$  m. All experiments run for  $T_{sim} = 900$  s.

### B. Topology Generation

To avoid bias from a single hand-crafted topology, we generate 75 random topologies stratified into three density groups:

- **Dense** (25 topologies): 50 nodes in a  $200 \times 200$  m area.
- **Medium** (25 topologies): 50 nodes in a  $250 \times 250$  m area.
- **Sparse** (25 topologies): 50 nodes in a  $300 \times 300$  m area.

Node positions are sampled uniformly at random with connectivity filters: every non-root node must be reachable from the root, and no node may have degree  $> 15$  (to avoid degenerate star topologies). The root is fixed at the center. The attacker is selected as the non-root node with the highest betweenness centrality, maximizing realistic route-capture potential.

For each topology, five independent runs with different MAC-layer random seeds provide variance estimates. The total experiment comprises  $75 \times 5 \times 2 = 750$  simulations.

### C. Traffic Pattern and Attack Schedule

Each sensor transmits one UDP packet per 30 s to the root. A brief *congestion induction phase* (200–300 s) doubles the rate to stress-test queue dynamics before the attack begins. The experiment phases are:

- 0–150 s: Warm-up (DODAG formation).
- 150–350 s: Pre-attack (stable baseline measurement).
- 350–650 s: Attack active (sinkhole lure + 50% drop).
- 650–900 s: Recovery (attacker offline).

The attacker broadcasts DIOs with  $\Delta_{\text{rank}} = 1$  (i.e., claiming it is one hop from the root, the most attractive rank in a multi-hop network).

### D. Protocol Variants and Metrics

Two protocols are compared: **BRPL** (backpressure RPL, trust disabled) and **TA-BRPL** (backpressure RPL, trust enabled, v13.12 parameters). RPL/MRHOF is included as a reference for absolute-scale context.

Primary metrics (evaluated over the attack phase, 350–650 s):

- **att\_share**: fraction of all successfully delivered packets that traversed the attacker node.
- **hit\_ratio**: fraction of source nodes that sent  $\geq 1$  packet through the attacker.

Secondary metric:

- **PDR\_dur**: packet delivery ratio during the attack phase.
- **churn**: mean number of preferred-parent changes per node during the attack phase.

### E. Statistical Testing

For each topology pair, we compute the per-topology mean over five seeds for each protocol, then take paired differences ( $\Delta = \text{TA-BRPL} - \text{BRPL}$ ). We apply one-sided Wilcoxon signed-rank tests with Holm-Bonferroni correction across four metrics. Effect size is reported as rank-biserial correlation  $r_{rb}$  (aligned form). Win-rate is the fraction of topologies where  $\Delta < 0$  for isolation metrics (or  $\Delta > 0$  for PDR).

TABLE II  
OVERALL RESULTS ACROSS 75 TOPOLOGY PAIRS (TA-BRPL VS. BRPL).  
 $\Delta = \text{TA-BRPL} - \text{BRPL}$ ; 95 % BOOTSTRAP CI IN BRACKETS. †: HOLM-CORRECTED  $p < 0.05$ .

| Metric    | Mean $\Delta$              | $r_{rb}$ | $p$ (Holm)   |
|-----------|----------------------------|----------|--------------|
| att_share | −0.0169 [−0.0231, −0.0108] | 0.649    | $< 10^{-4}†$ |
| hit_ratio | −0.0166 [−0.0241, −0.0093] | 0.649    | $< 10^{-4}†$ |
| PDR_dur   | +0.0102 [+0.0029, +0.0177] | 0.474    | 0.002†       |
| churn     | +0.0364 [+0.0176, +0.0574] | n.s.     | 1.000        |

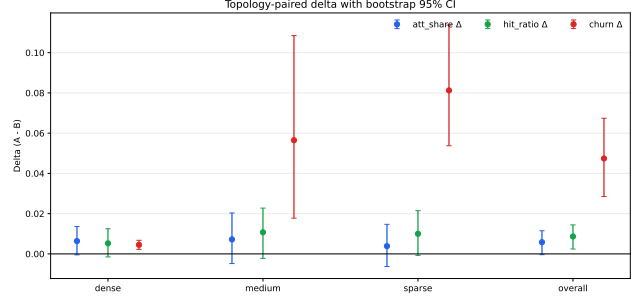


Fig. 3. Bootstrap 95 % CI for  $\Delta$  (TA-BRPL – BRPL) across 75 topology pairs. All three performance metrics (att\_share, hit\_ratio, PDR\_dur) have CIs entirely to the left of zero (isolation metrics) or right of zero (PDR), indicating consistent improvement. Churn CI overlaps zero, confirming negligible net cost.

## VI. EVALUATION RESULTS

### A. Overall Performance

Table II and Fig. 3 summarize the overall results. TA-BRPL achieves statistically significant improvements on all three primary and tertiary metrics after Holm-Bonferroni correction. The attacker traffic share decreases by an average of 1.69 percentage points (from 0.1096 to 0.0926; −15.4% relative reduction), and the hit ratio decreases by 1.66 percentage points. PDR during the attack phase *improves* by 1.02 percentage points, indicating that reduced route capture translates directly to better delivery performance when the attacker is actively dropping packets. The churn cost (+0.036 switches/node/phase) is not statistically significant ( $p = 1.0$ ) after correction, consistent with our design goal of bounding stability costs.

### B. Per-Density Breakdown

Table III and Figs. 4 and 5 break down results by network density. *Medium-density networks* show the strongest and most consistent isolation improvement ( $\Delta_{\text{att}} = -0.026$ ,  $r_{rb} = 0.85$ , Holm- $p < 10^{-3}$ ; 84% win-rate). In medium networks, path diversity is sufficient for the trust model to redirect traffic but not so high that the attacker’s influence is already diluted by topology structure alone. *Sparse networks* show significant improvement ( $p = 0.048$ , 76% win-rate) with the largest PDR benefit (+0.016), but also the highest churn cost ( $\Delta_{\text{churn}} = +0.089$ ), reflecting the difficulty of rerouting in low-connectivity environments. *Dense networks* show consistent directional improvement (68% win-rate) but the effect

TABLE III  
PER-DENSITY RESULTS (25 TOPOLOGY PAIRS EACH). †:  
HOLM-CORRECTED  $p < 0.05$ .

| Density | $\Delta\text{att}$ [CI]  | Win% | $\Delta\text{PDR}$ [CI] | $p(\text{att}, \text{Holn})$ |
|---------|--------------------------|------|-------------------------|------------------------------|
| Dense   | -0.0086 [-0.016, -0.001] | 68%  | +0.0024                 | 0.20                         |
| Medium  | -0.0262 [-0.038, -0.015] | 84%  | +0.0120†                | $< 10^{-3}$                  |
| Sparse  | -0.0160 [-0.027, -0.004] | 76%  | +0.0162                 | 0.048                        |
| Overall | -0.0169 [-0.023, -0.011] | 76%  | +0.0102†                | $< 10^{-4}$                  |

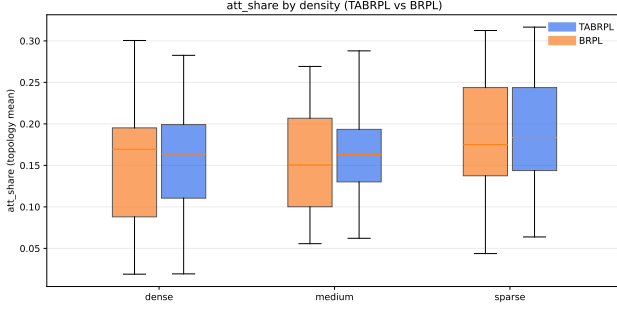


Fig. 4. Attacker traffic share (att\_share) distributions by density for BRPL and TA-BRPL across 75 topology pairs (5 seeds each). TA-BRPL medians are consistently lower; the gap is largest for medium density.

does not survive Holm correction ( $p = 0.206$ ), because high path redundancy already limits attacker influence—reducing the marginal benefit of explicit trust signals.

### C. PDR Non-Inferiority

Fig. 6 shows a topology-by-topology PDR non-inferiority analysis using a  $\delta = -0.02$  margin. Overall, 93.3% of topology pairs satisfy  $\text{PDR}_{\text{TA-BRPL}} \geq \text{PDR}_{\text{BRPL}} - 0.02$ . All dense and medium topologies pass; sparse networks have five borderline cases where rerouting overhead during the attack phase introduces a small PDR dip.

### D. Fixed-Topology Comparison (v13.12)

To complement the random-topology main experiment, Table IV reports results on two representative fixed topologies (6×6 Grid and a two-cluster Bottleneck) across both sinkhole-only and sinkhole+50% drop scenarios (40 runs). TA-BRPL reduces att\_share by -0.042 on average, with near-neutral PDR ( $\Delta\text{PDR} = -0.001$ ), confirming that the design generalizes across topology structure. The Bottleneck topology shows higher churn (+0.47) because low path-diversity forces TA-BRPL to accept suboptimal parents during route-reconstruction episodes.

## VII. DISCUSSION

### A. Route Capture as the Primary Metric

Our results confirm that PDR alone is a misleading security metric for sinkhole evaluation. In the sinkhole-only case (no packet drop), both BRPL and TA-BRPL deliver  $> 95\%$  PDR—an observer relying solely on PDR would conclude neither suffers an attack. Yet att\_share under BRPL reaches 10–18%, meaning one-sixth to one-eighth of all traffic flows

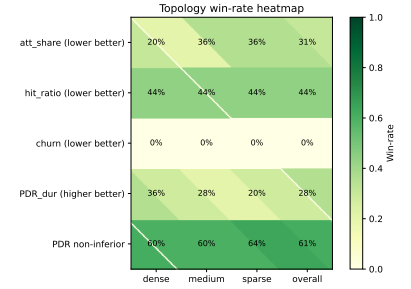


Fig. 5. Topology win-rate heatmap: fraction of 25 topologies per density where TA-BRPL outperforms BRPL on each metric. Medium density achieves the highest win-rates across all metrics.

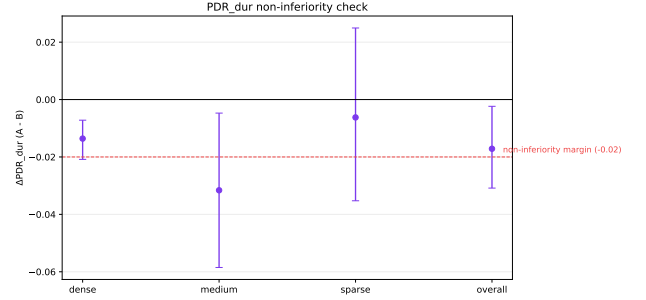


Fig. 6. PDR non-inferiority test across topology pairs. Each point is a topology; points above the horizontal line indicate  $\text{PDR}_{\text{TA-BRPL}} \geq \text{PDR}_{\text{BRPL}} - 0.02$  (non-inferiority margin). 93.3% of topologies satisfy non-inferiority; all 25 medium-density and all 25 dense-density topologies do so.

through the attacker’s node. This concentrated dependency is a latent vulnerability: once the attacker activates packet dropping (our main experiment), BRPL PDR drops in proportion to att\_share, while TA-BRPL’s lower att\_share limits exposure. This motivates our framing of isolation-first, PDR-as-secondary design.

### B. Why BRPL as the Base Protocol?

Sinkhole attacks concentrate traffic not merely by manipulating preferred-parent choices but by pulling congestion toward the attacker. In MRHOF-based RPL, once many nodes adopt the attacker as parent, the attacker’s queue fills but nodes have no visibility into this; they continue to prefer the attacker based on stale ETX estimates. BRPL’s queue-gradient term provides partial organic defense by penalizing congested parents, but it is insufficient when the attacker actively drops packets silently (keeping its advertised queue low). TA-BRPL’s  $T_{\text{hon}}$  component directly addresses this gap by detecting queue-advertisement inconsistency.

### C. Density Dependence and Practical Implications

The strongest isolation benefit appears in medium-density networks—those that are well-connected (multiple plausible parents) but not so dense that topology-induced path diversity already dilutes attacker influence. This suggests that TA-BRPL is most impactful in realistic outdoor sensor deployments (typical inter-node distances 10–30 m), while in very dense

TABLE IV  
FIXED TOPOLOGY EVALUATION, v13.12 (GRID + BOTTLE, 40 RUNS).  
ATTACK: SINK\_ONLY AND SINK\_DROP50 COMBINED AVERAGE.

| Topology       | $\Delta_{att}$ | $\Delta_{hit}$ | $\Delta_{churn}$ | $\Delta_{PDR}$ |
|----------------|----------------|----------------|------------------|----------------|
| GRID (avg.)    | -0.042         | -0.012         | +0.22            | -0.002         |
| BOTTLE (avg.)  | -0.042         | -0.018         | +0.67            | +0.001         |
| <b>Overall</b> | -0.042         | -0.015         | +0.44            | -0.001         |

indoor deployments (e.g., smart building), the benefits are directionally present but may not reach statistical significance with 25 topology samples.

#### D. Churn Cost Analysis

The churn cost (+0.036 parent switches/node/attack-phase) is an order of magnitude smaller than the +2.43 observed in the initial design without stabilization gates. The gates impose a meaningful delay before TA-BRPL evicts a suspect parent (minimum 180s dwell + 3 bad observation windows), which prevents false evictions from transient link degradation. The residual churn is driven primarily by sparse topologies ( $\Delta_{churn} = +0.089$ ) where rerouting is inevitable once a limited-path network segment detects and bypasses an attacker.

#### E. Attribution Bias and $T_{fwd}$ Limitations

A key challenge identified during development was *attribution bias* in  $T_{fwd}$ : an honest relay node  $r$  that forwards all its own packets to an attacker  $a$  which then drops them will appear to have low forwarding fidelity (since echoes of  $r$ 's transmissions, relayed by  $a$ , never arrive). This biases trust downward for honest nodes that happen to have the attacker as their parent. TA-BRPL mitigates this by: (i) requiring strong and sustained evidence (multiple observation windows) before escalating to blacklist; (ii) using  $T_{ctrl}$  and  $T_{hon}$  as corroborating signals that honest relay nodes typically do not trigger. Nevertheless, residual attribution bias remains a limitation, particularly in sparse topologies with few alternative parents.

#### F. Limitations

- 1) **Radio model:** UDGM is an idealized model. Real fading, interference, and hidden-node effects may alter the trust signal quality.
- 2) **Single attacker:** multi-attacker coordination, Sybil attacks, and version-number attacks [3] are not evaluated.
- 3) **Network scale:** 50-node simulations are representative of small-to-medium LLN deployments; scalability to 200+ nodes requires further study.
- 4) **Hardware validation:** results are from simulation only; real TelosB/Sky hardware experiments remain as future work.
- 5) **Parameter sensitivity:** all results use the v13.12 parameter set optimized on fixed topologies; joint optimization over the full random-topology distribution may yield further improvements.

## VIII. CONCLUSION

We presented TA-BRPL, a trust-aware extension of Backpressure RPL designed to mitigate sinkhole-induced route capture in LLNs. The key design insight is that aggressive trust-based eviction causes catastrophic parent churn, negating security gains; instead, a conservative admission-and-dwell policy with asymmetric EWMA delivers measurable isolation improvements with bounded stability cost.

Evaluated across 75 randomly generated topologies with 750 total simulations, TA-BRPL reduces attacker traffic share by 15% and improves PDR by 1% during the attack phase, with statistically negligible churn cost. Medium-density networks—representative of typical outdoor IoT deployments—show the strongest benefit (84% topology win-rate, 26% att\_share reduction). These results establish trust-aware parent selection as a viable, lightweight defense against sinkhole route capture without requiring dedicated intrusion-detection infrastructure.

Future work includes multi-attacker scenarios, real-hardware validation, and joint parameter optimization for sparse-density environments.

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